Zhongrui Wang

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Education

University of Wisconsin-Madison — Madison, WI
- *Ph.D. Candidate - Mathematics (Minor: Computer Sciences), Advisor: Nan Chen* — *Jul 2023 – present*
  M.A. - Mathematics — *Aug 2026*

Nanjing University — Nanjing, China
- *M.S. - Atmospheric Science, Advisor: Lili Lei* — *Sep 2020 – Jun 2023*
  B.S. - Atmospheric Science — *Sep 2016 – Jun 2020*

Research Interests

My research lies at the interface of applied and computational mathematics, machine learning, and geophysics. I draw research questions from geophysical problems, such as weather and climate prediction, geophysical turbulence, and the El Niño–Southern Oscillation (ENSO), and develop data assimilation, uncertainty quantification, and scientific machine learning methods that return to these applications. Current topics:

- **Closed-form nonlinear filtering** for high-dimensional, partially observed stochastic systems, without Monte Carlo sampling.
- **Learning tractable dynamics**: conditional Gaussian Koopman networks that make nonlinear dynamics analytically tractable for data assimilation.
- **Regime-dependent prediction**: online regime inference with adaptive weighting of multiple models.
- **Digital twins**: an ENSO twin that embeds a conceptual model in a conditional Gaussian Koopman network, with closed-form assimilation and centuries-stable runs.
- **Machine learning for ensemble data assimilation**: CNN-based adaptive covariance localization.

Publications

- **Zhongrui Wang** and Nan Chen. "A Physics-Embedded Conditional Gaussian Koopman Network as a Digital Twin of ENSO with Closed-Form Data Assimilation" (in preparation), 2026
- **Zhongrui Wang** and Nan Chen. "Discrete- and Continuous-Time Data Assimilation: The Role of Observational Noise and Bridges" (in preparation), 2026
- **Zhongrui Wang**, Rebecca Willett, Nan Chen, and Yinling Zhang. "A Regime-Dependent Multi-Model Framework for Data Assimilation and Prediction", *Quarterly Journal of the Royal Meteorological Society* (submitted), 2026
- **Zhongrui Wang**, Chuanqi Chen, Jin-Long Wu, and Nan Chen. "A Lagrangian Conditional Gaussian Koopman Network for Data Assimilation and Prediction", *Journal of Computational Physics* (under review), 2026
- Chuanqi Chen, **Zhongrui Wang**, Nan Chen, and Jin-Long Wu. "Modeling partially observed nonlinear dynamical systems and efficient data assimilation via discrete-time conditional Gaussian Koopman network", *Computer Methods in Applied Mechanics and Engineering* 445, 118189, 2025.
- **Zhongrui Wang**, Nan Chen, and Di Qi. "A Nonlinear Data Assimilation Algorithm with Closed-Form Approximations for Multilayer Flow Fields", *Monthly Weather Review* 153, 12, 2889–2912, 2025.
- **Zhongrui Wang**, Haohao Sun, Lili Lei, Zhe-Min Tan, and Yi Zhang. "The importance of data assimilation components for initial conditions and subsequent error growth", *Science China Earth Sciences*, 2024.
- **Zhongrui Wang**, Lili Lei, Jeffrey L. Anderson, Zhe-Min Tan, and Yi Zhang. "Convolutional Neural Network–Based Adaptive Localization for an Ensemble Kalman Filter", *Journal of Advances in Modeling Earth Systems* 15, 10, e2023MS003642, 2023.
- Lili Lei, **Zhongrui Wang**, and Zhe-Min Tan. "Integrated Hybrid Data Assimilation for an Ensemble Kalman Filter", *Monthly Weather Review* 149, 12, 4091-4105, 2021.
- Shibao Wang, Yun Ma, **Zhongrui Wang**, et al. "Mobile monitoring of urban air quality at high spatial resolution by low-cost sensors: impacts of COVID-19 pandemic lockdown", *Atmospheric Chemistry and Physics* 21, 7199–7215, 2021.
- Yanxu Zhang, . . . , **Zhongrui Wang**, et al. "Large-eddy simulation of traffic-related air pollution at a very high resolution in a mega-city: evaluation against mobile sensors and insights for influencing factors", *Atmospheric Chemistry and Physics* 21, 2917–2929, 2021.

Conferences and Seminars

- **Poster**: The University of Chicago and Caltech Conference on AI+Science, Chicago, Oct 2026
- **Poster**: IMSI: Unpacking AI Weather Emulators, Chicago, USA, Sep-Oct, 2026
- **Talk**: Fudan University, Shanghai, China, Sep 2026
- **Talk**: Shanghai University of Finance and Economics, Shanghai, China, Sep 2026
- **Talk and Poster**: 8th WMO Symposium on Data Assimilation & 12th International Symposium on Data Assimilation (ISDA), Nanjing, Aug-Sep 2026
- **Poster**: IFDS Research Expo, Institute for Foundations of Data Science, UW–Madison, Madison, Aug 2026
- **Poster**: AI+Science Summer School, University of Chicago, Chicago, Jun 2026
- **Talk**: SIAM Conference on Uncertainty Quantification (UQ26), Minneapolis, Mar 2026
- **Poster**: IMSI: Data Assimilation and Inverse Problems for Digital Twins, Chicago, Oct 2025
- **Talk**: Scientific Machine Learning: Theory, Algorithms, and Applications, Purdue University, Sep 2025
- **Poster**: Data Meets Dynamics: Workshop on Data Assimilation for Complex Systems and Applications, Madison, Aug 2025
- **Talk**: Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, Jul-Aug 2025
- **Talk**: SIAM Conference on Applications of Dynamical Systems (DS25), minisymposium on nonlinear and stochastic dynamics in weather and climate science, Denver, May 2025
- **Talk**: The Institute for Foundations of Data Science (IFDS) ideas forum, Madison, May 2025
- **Talk**: ISDA Online: Nonlinear Data Assimilation, Online, May 2025
- **Poster**: The American Geophysical Union (AGU) Annual Meeting, Washington, D.C., Dec 2024
- **Talk**: 102nd American Meteorological Society (AMS) Annual Meeting, 26th Conference on Integrated Observing and Assimilation Systems for the Atmosphere, Oceans, and Land Surface (IOAS-AOLS), Jan 2022

Academic Service

- Reviewer: Physica D: Nonlinear Phenomena, 2026
- Reviewer: npj Artificial Intelligence, 2025

Teaching

- **Teaching Assistant** — University of Wisconsin-Madison
  Math 211: Survey of Calculus I — *Fall 2025, Spring 2026*
- **Undergraduates Mentoring** — University of Wisconsin-Madison
  Madison Experimental Mathematics Lab (MXM) — *Spring 2025*
  Project: Integrating Machine Learning with Data Assimilation for Complex Dynamical System Predictions
- **Teaching Assistant** — University of Wisconsin-Madison
  Math 114: Precalculus — *Fall 2023*
- **Teaching Assistant** — Nanjing University
  Dynamic Meteorology — *Fall 2021*

Awards

- SIAM Student Travel Award, SIAM Conference on Uncertainty Quantification (UQ26), 2026
- SIAM Student Travel Award, Third Joint SIAM/CAIMS Annual Meetings (AN25), 2025
- UW–Madison Student Graduate Research Grant, 2025
- People's Scholarship, Nanjing University, 2017, 2018, 2019

Programming

Python (NumPy, SciPy, pandas, PyTorch, JAX, xarray), MATLAB, R, Git, Linux